TYPES OF MALWARE

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WHAT IS MALWARE

- Malware is short for malicious software.
- It refers to any software or code intentionally designed to harm, exploit, disrupt, or gain unauthorized access to a computer system, network, or data.
- Malware is typically created to:
  - Steal sensitive data (passwords, bank details, PHI, company data)
  - Spy on users (keylogging, screen capture)
  - Damage or delete files
  - Lock systems and demand money (ransom)
  - Take control of computers (botnets)
  - Disrupt business operations

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1. Stealer Malware (Information Stealer)

- Stealer malware is designed to silently collect and steal sensitive information from a victim's device and send it to the attacker.

- It commonly targets:
  - Saved browser passwords
  - Banking and financial information
  - Cookies and session tokens
  - Cryptocurrency wallet data
  - Email credentials
  - System information

- Many modern stealers (like RedLine Stealer, Vidar Stealer, and Raccoon Stealer) are sold on the dark web as Malware-as-a-Service (MaaS), making them widely used by cybercriminals.

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TYPES OF MALWARE

How it spreads:

- Phishing emails with malicious attachments
- Fake software downloads or cracked software
- Malicious browser extensions
- Infected websites and drive-by downloads

Ways to protect against it:

- Avoid downloading pirated/cracked software.
- Use updated antivirus/EDR solutions to detect stealer activity.
- Enable Multi-Factor Authentication (MFA) for important accounts.
- Keep browsers and operating systems updated.
- Do not save sensitive passwords in browsers without protection.
- Be cautious with email attachments and unknown links.

Key risk:

Even if the malware is removed, stolen credentials may already be compromised, so changing passwords immediately is important after infection.

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2. RAT (Remote Access Trojan)

- A RAT is a type of malware that allows an attacker to remotely control an infected computer without the user's knowledge.

- Once installed, the attacker can:

  - Access, modify, or delete files
  - Monitor user activity
  - Capture screenshots or record keystrokes
  - Turn on the webcam or microphone
  - Install additional malware

- Common examples include DarkComet RAT, njRAT, and NanoCore RAT.

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How it spreads:

- Phishing email attachments
- Fake software or cracked applications
- Malicious downloads from untrusted websites
- Infected USB drives

Ways to protect against it:

- Do not open suspicious email attachments or links.
- Download software only from trusted sources.
- Use updated antivirus/endpoint security tools.
- Keep operating systems and applications patched.
- Enable firewalls to monitor unauthorized remote connections.
- Restrict admin privileges to prevent unauthorized installations.

Key risk:

RATs give attackers full remote control of the system, which can lead to data theft, espionage, and further malware infections..

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3. Loader Malware

- A Loader malware is designed to download and install other types of malware onto an infected system. Its main role is to prepare the system for further attacks by silently delivering additional malicious payloads such as ransomware, spyware, or trojans.

- Well-known loader malware includes Emotet, BazarLoader, and SmokeLoader.

How it spreads:

- Phishing emails with malicious attachments or links
- Infected documents with macros (Word/Excel)
- Fake software downloads
- Exploit kits on compromised websites

Ways to protect against it:

- Disable macros in Office documents from unknown sources.
- Use updated antivirus/EDR solutions to detect suspicious downloads.
- Avoid opening attachments from unknown emails.
- Keep operating systems and software patched.
- Implement email filtering and web security controls.

Key risk: Loader malware itself may seem harmless initially, but it opens the door for more dangerous malware, such as ransomware or banking trojans, making it a major entry point in modern cyberattacks.

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4. Ransomware

- Ransomware is a type of malware that encrypts files or locks a computer system, preventing the user from accessing their data. The attacker then demands a ransom payment (usually in cryptocurrency) to provide a decryption key to restore the files.

- Well-known ransomware families include WannaCry, LockBit, and Ryuk ransomware.

How it spreads:

- Phishing emails with malicious attachments or links
- Exploiting unpatched software vulnerabilities
- Infected websites or downloads
- Remote Desktop Protocol (RDP) attacks
- Loader malware or trojans

Ways to protect against it:

- Maintain regular offline backups of important data.
- Keep operating systems and software updated.
- Use antivirus/endpoint protection tools.
- Be cautious with email attachments and links.
- Disable unused RDP access or secure it with strong authentication.
- Use network segmentation and least-privilege access.

Key risk: Ransomware can cause data loss, business disruption, and financial damage, especially if backups are not available. Many modern attacks also steal data before encryption and threaten to leak it if the ransom is not paid.

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5. Botnet Malware

- Botnet malware infects computers or devices and turns them into “bots” (zombie machines) that are remotely controlled by attackers. A group of these infected devices forms a botnet, which attackers use to perform large-scale cyberattacks such as:

  - Distributed Denial-of-Service (DDoS) attacks
  - Sending spam emails
  - Spreading more malware
  - Cryptocurrency mining
  - Credential-stuffing attacks

- Famous botnets include Mirai botnet, Zeus malware, and Gameover Zeus.

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How it spreads:

- Infected email attachments
- Malicious downloads or cracked software
- Exploiting vulnerabilities in systems or IoT devices
- Weak passwords on network-connected devices

Ways to protect against it:

- Use strong and unique passwords, especially for network devices.
- Keep operating systems, software, and IoT devices updated.
- Install antivirus and endpoint security solutions.
- Avoid downloading software from untrusted sources.
- Monitor network traffic for unusual activity.

Key risk:

Infected devices may work normally but are secretly controlled by attackers, who can use them to launch large-scale cyberattacks without the owner's knowledge.

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6. Backdoor Malware

- Backdoor malware creates a hidden method to bypass normal authentication and security controls, allowing attackers to access a computer or network secretly. Once installed, attackers can:
  - Gain unauthorized remote access
  - Steal sensitive data
  - Install additional malware
  - Monitor or control system activities
- Examples of malware associated with backdoor capabilities include Back Orifice, Poison Ivy RAT, and DoublePulsar.

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How it spreads:

- Trojanized software or fake downloads
- Phishing emails with malicious attachments
- Exploiting system vulnerabilities
- Installed by other malware (like trojans or loaders)

Ways to protect against it:

- Keep systems and applications patched and updated.
- Use antivirus/EDR tools to detect hidden access.
- Avoid downloading software from untrusted sources.
- Monitor network traffic for unusual remote connections.
- Use firewalls and strong access control policies.

Key risk:

Backdoor malware allows attackers to maintain long-term hidden access to systems, making it dangerous for data theft, espionage, and persistent cyberattacks.

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7. Keylogger

- A keylogger is malware that records every keystroke typed on a keyboard and secretly sends the data to an attacker. This allows attackers to capture:
  - Usernames and passwords
  - Banking and credit card information
  - Emails and private messages
  - Other sensitive data entered on the computer
- Keylogging functionality is often found in malware such as Agent Tesla, HawkEye Keylogger, and Ardamax Keylogger.

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How it spreads:

- Phishing email attachments
- Malicious downloads or cracked software
- Trojan malware installations
- Infected websites or fake software updates

Ways to protect against it:

- Use updated antivirus or endpoint protection software.
- Avoid opening suspicious email attachments or links.
- Download software only from trusted sources.
- Enable Multi-Factor Authentication (MFA) for accounts.
- Keep operating systems and applications updated

Key risk:

Keyloggers can silently capture login credentials and financial information, leading to identity theft and unauthorized access to accounts.

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8. Adware

- Adware is malware that automatically displays unwanted advertisements on a user's device, usually through pop-ups, banners, or redirects to advertising websites. It may also track browsing activity to show targeted ads and sometimes slow down the system.

- Examples of adware-related threats include Fireball malware, Gator adware, and DollarRevenue adware..

How it spreads:

- Bundled with free software downloads
- Fake software updates
- Malicious browser extensions
- Infected websites or pop-up ads

Ways to protect against it:

- Download software only from trusted sources.
- Avoid clicking suspicious ads or pop-ups.
- Use ad blockers and updated antivirus software.
- Check installation options and avoid bundled programs.
- Regularly scan and remove unwanted browser extensions.

Key risk: Although adware mainly shows ads, it can collect browsing data, slow down systems, and sometimes lead to more serious malware infections.

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9. Trojan (Trojan Horse)

- A Trojan is malware that disguises itself as legitimate software to trick users into installing it. Once installed, it can:
  - Steal sensitive information
  - Install additional malware
  - Create backdoors for attackers
  - Monitor user activity or damage files
- Examples include Zeus Trojan, Emotet, and TrickBot.

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TYPES OF MALWARE

How it spreads:

- Fake or cracked software downloads
- Phishing email attachments
- Malicious links or websites
- Infected USB drives

Ways to protect against it:

- Download software only from trusted websites.
- Do not open suspicious email attachments or links.
- Use updated antivirus or endpoint protection.
- Keep operating systems and software patched.
- Avoid using pirated or cracked software.

Key risk:

Trojans can silently install other malware and give attackers unauthorized access, making them a common entry point for many cyberattacks.

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TYPES OF MALWARE

10. Crypto Miner Malware (Cryptojacking Malware)

- rypto miner malware secretly uses the victim’s computer resources (CPU/GPU) to mine cryptocurrency for the attacker without the user’s permission. This can cause:
  - Slow system performance
  - Overheating of devices
  - Increased electricity usage
  - Reduced hardware lifespan
- Examples include Coinhive, Smominru, and WannaMine.

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TYPES OF MALWARE

How it spreads:

- Fake or cracked software downloads
- Phishing email attachments
- Malicious links or websites
- Infected USB drives

Ways to protect against it:

- Download software only from trusted websites.
- Do not open suspicious email attachments or links.
- Use updated antivirus or endpoint protection.
- Keep operating systems and software patched.
- Avoid using pirated or cracked software.

Key risk:

Trojans can silently install other malware and give attackers unauthorized access, making them a common entry point for many cyberattacks.

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11. InfoStealer Malware

- InfoStealer malware is designed to steal sensitive information from an infected system and send it to attackers. It commonly targets:
  - Browser passwords and cookies
  - Banking and financial data
  - Email credentials
  - Cryptocurrency wallet information
  - System and application data
- Examples include RedLine Stealer, Vidar Stealer, and Raccoon Stealer.

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TYPES OF MALWARE

How it spreads:

- Phishing emails with malicious attachments
- Fake software downloads or cracked software
- Malicious browser extensions
- Bundled with other malware like trojans or loaders

Ways to protect against it:

- Avoid downloading pirated or unknown software.
- Use updated antivirus or endpoint security tools.
- Enable Multi-Factor Authentication (MFA) for important accounts.
- Keep operating systems and browsers updated.
- Be cautious with email attachments and suspicious links.

Key risk:

InfoStealers can silently steal login credentials and financial data, which can lead to account compromise, identity theft, and financial fraud.

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TYPES OF MALWARE

12. Spyware

- Spyware is malware designed to secretly monitor and collect information about a user's activities without their knowledge. It can track:
  - Browsing history
  - Login credentials
  - Emails and messages
  - Personal and financial information
- Examples of spyware include Pegasus spyware, CoolWebSearch, and FinFisher.

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TYPES OF MALWARE

How it spreads:

- Malicious software downloads
- Fake applications or browser extensions
- Phishing emails and attachments
- Infected websites or bundled freeware

Ways to protect against it:

- Install trusted antivirus or anti-spyware software.
- Download applications only from official sources.
- Avoid clicking suspicious links or attachments.
- Keep operating systems and applications updated.
- Regularly scan devices for malware.

Key risk:

Spyware operates silently in the background, allowing attackers to collect personal and confidential information without the user noticing.

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TYPES OF MALWARE

13. Banking Trojan

- A Banking Trojan is malware specifically designed to steal online banking credentials and financial information. It often works by:
  - Capturing bank login usernames and passwords
  - Performing keylogging
  - Creating fake banking pages (web injection) to capture details
  - Intercepting one-time passwords (OTP) or transaction data
- Common banking trojans include Zeus Trojan, Dridex, and TrickBot.

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TYPES OF MALWARE

How it spreads:

- Phishing emails pretending to be from banks
- Malicious attachments (Word, Excel, PDFs)
- Fake banking or financial apps
- Infected downloads and exploit kits

Ways to protect against it:

- Avoid clicking links in suspicious banking emails.
- Use official banking websites or apps only.
- Enable Multi-Factor Authentication (MFA).
- Keep antivirus and operating systems updated.
- Regularly monitor bank transactions for unusual activity.

Key risk:

Banking trojans can steal financial credentials and perform fraudulent transactions, leading to direct financial loss for individuals and organizations.

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14. Wiper Malware

- Wiper malware is designed to permanently delete or destroy data on a computer or network, making systems unusable. Unlike ransomware, it does not aim to recover files, but instead causes maximum damage by erasing files, system data, or even the operating system.

- Notable examples include Shamoon, NotPetya, and WhisperGate.

How it spreads:

- Phishing emails with malicious attachments
- Exploiting unpatched software vulnerabilities
- Compromised networks or supply chain attacks
- Installed by other malware like trojans or loaders

Ways to protect against it:

- Maintain regular offline backups of important data.
- Keep operating systems and software updated.
- Use antivirus/endpoint detection tools.
- Restrict network access and apply least-privilege policies.
- Monitor networks for suspicious activities.

Key risk: Wiper malware can cause complete and irreversible data loss, leading to major operational disruption and financial damage for organizations.

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15. Fileless Malware

- Fileless malware operates directly in a computer's memory (RAM) instead of installing malicious files on the hard drive. It often uses legitimate system tools to execute attacks, making it difficult for traditional antivirus software to detect.

- Common examples include attacks associated with PowerShell Empire, Poweliks, and Kovter.

How it spreads:

- Phishing emails with malicious scripts
- Exploiting software vulnerabilities
- Malicious macros in Office documents
- Compromised websites or drive-by downloads

Ways to protect against it:

- Disable macros in Office documents from unknown sources.
- Keep operating systems and software patched.
- Use advanced endpoint detection and response (EDR) tools.
- Monitor system processes and PowerShell activity.
- Limit administrative privileges and enforce application control.

Key risk: Because fileless malware does not leave traditional files on the system, it can bypass signature-based security tools and remain hidden while executing attacks.

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16. Dropper Malware

- Dropper malware is designed to install or “drop” other malicious software onto a victim’s system. Its main purpose is to deliver and secretly install additional malware such as trojans, ransomware, or spyware while avoiding detection.

- Examples include Zlob Trojan, Emotet, and SmokeLoader.

How it spreads:

- Phishing emails with infected attachments
- Fake software downloads or cracked programs
- Malicious websites or exploit kits
- Bundled with seemingly legitimate applications

Ways to protect against it:

- Avoid downloading software from untrusted sources.
- Do not open suspicious email attachments or links.
- Use updated antivirus or endpoint protection tools.
- Keep operating systems and applications patched.
- Disable macros in Office documents from unknown senders.

Key risk: Dropper malware acts as a delivery mechanism for more dangerous malware, allowing attackers to infect systems with multiple threats without the user noticing.

TYPES OF MALWARE

17. Downloader Malware

- Downloader malware is designed to connect to a remote server and download additional malicious programs onto an infected system. It usually acts as a first-stage infection that brings in other malware like ransomware, spyware, or trojans.

- Examples include Upatre, BazarLoader, and SmokeLoader.

How it spreads:

- Phishing emails with malicious attachments
- Fake software updates or downloads
- Infected websites and exploit kits
- Bundled with pirated or cracked software

Ways to protect against it:

- Avoid opening suspicious email attachments or links.
- Download software only from trusted and official sources.
- Use updated antivirus or endpoint protection tools.
- Keep operating systems and applications patched.
- Enable web and email security filtering.

Key risk: Downloader malware may appear harmless initially, but it can bring multiple dangerous malware infections into the system, making it a common entry point in cyberattacks.

TYPES OF MALWARE

18. Rootkit

- A rootkit is malware designed to gain and maintain privileged (root or administrator) access to a system while hiding its presence. It can:
  - Hide files, processes, or other malware
  - Give attackers full control of the system
  - Disable security tools
  - Allow long-term stealthy access to the device
- Examples include ZeroAccess rootkit, TDSS rootkit, and Sony BMG rootkit.

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TYPES OF MALWARE

How it spreads:

- Infected software downloads or pirated programs
- Phishing email attachments
- Exploiting system vulnerabilities
- Installed by trojans or other malware

Ways to protect against it:

- Keep operating systems and software updated.
- Use advanced antivirus or anti-rootkit tools.
- Avoid downloading software from untrusted sources.
- Use secure boot and system integrity checks.
- Regularly monitor systems for unusual behavior.

Key risk:

Rootkits are very difficult to detect and remove because they hide deep within the system, allowing attackers to maintain persistent and stealthy control over the device.

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TYPES OF MALWARE

19. Mobile Malware

- Mobile malware is malicious software designed to infect smartphones or tablets and perform harmful activities such as:
  - Stealing personal and financial information
  - Tracking location and user activity
  - Sending premium SMS messages
  - Installing additional malware
  - Spying through the camera or microphone
- Examples include Joker malware, FluBot, and Pegasus spyware.

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TYPES OF MALWARE

How it spreads:

- Downloading apps from unofficial app stores
- Malicious mobile apps disguised as legitimate ones
- SMS phishing (smishing) links
- Infected websites or malicious advertisements

Ways to protect against it:

- Download apps only from official stores like the Google Play Store or Apple App Store.
- Avoid clicking suspicious SMS or email links.
- Keep mobile OS and apps updated.
- Use mobile security or antivirus apps.
- Review app permissions before installing.

Key risk:

Mobile malware can steal personal data, financial information, and sensitive communications, putting both individual users and organizations at risk.

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20. IoT Malware

- IoT malware targets Internet of Things (IoT) devices such as smart cameras, routers, smart TVs, and home automation systems. Once infected, attackers can:
  - Take remote control of the device
  - Use it in botnet attacks (DDoS)
  - Spy on users or collect data
  - Spread malware to other network devices
- Examples include Mirai botnet, Mozi botnet, and Gafgyt botnet..

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How it spreads:

- Weak or default passwords on IoT devices
- Unpatched device vulnerabilities
- Open network ports or insecure configurations
- Compromised firmware updates

Ways to protect against it:

- Change default usernames and passwords on IoT devices.
- Keep device firmware updated.
- Disable unnecessary services and ports.
- Use secure Wi-Fi networks and firewalls.
- Segment IoT devices on a separate network.

Key risk:

IoT malware can turn thousands of devices into large botnets used for massive cyberattacks, often without the device owner noticing.

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THANK YOU

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